

WASP-135b

R-1362

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-135_210623 · created 2026-07-31T23:47:47+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2459388.889 +/- 0.0057 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.174 +/- 0.051
Transit depth (Rp/Rs)^2	3.02 +/- 1.78 [%]
Orbital Inclination (inc)	82.95 +/- 2.86
Airmass coefficient 1 (a1)	14.58 +/- 9.45
Airmass coefficient 2 (a2)	-2.41 +/- 0.66
Scatter in the residuals of the lightcurve fit is	62.75 %
Best Comparison Star	None
Optimal Aperture	1.22
Optimal Annulus	4.5
Transit Duration (day)	0.072 +/- 0.022

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.174 ± 0.051 vs 0.139 (deviation 0.61σ)
Duration fit vs published	0.072 d vs 0.0687 d (deviation 0.13σ)
Mid-time O–C	12.2 min
Depth SNR	3.0
Residual scatter	62.75 %

Light curve

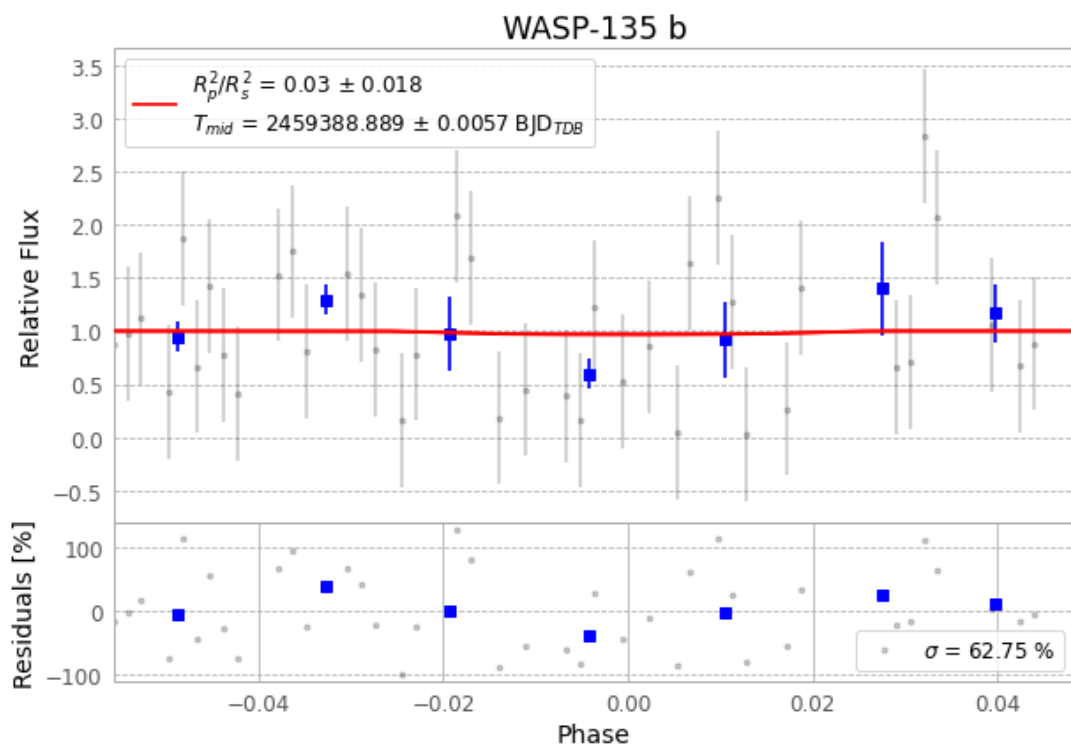


Figure 1 — FinalLightCurve_WASP-135 b_2021-06-23.png (SHA-256 02dd92ee2f0efdfc...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [619, 268], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 9201622d6b3a47ef...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

72 FITS frames (47 MB), WASP-135210623072112.FITS → WASP-135210623105412.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-135-210623.log (9d5f60632608...), FinalLightCurve_WASP-135 b_2021-06-23.png (02dd92ee2f0e...), FinalLightCurve_WASP-135 b_2021-06-23.csv (cb68ac23f73e...)

Compute resources

Wall time 260.5 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-31 23:47Z.